Power Budget

Team Number: 209
Project Name: **Automated Laundry Folding Robot**
Team Member Names: **Diego, Ruben, Zainan, Kristen**
Version: **1**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	DC Gearmotor	FIT0492-A	12 V	1	2,190.00	2,190.00	mA
	H-Bridge Motor Driver	DRV8876PWPR	4.5-37 V	1	7.00	7.00	mA
	PIC18F57Q43 Curiosity Nano	DM164150	1.8-5.5 V	1	100.00	100.00	mA
	Quad Op-Amp	MCP6004-I/P	1.8-6 V	1	0.68	0.68	mA
	Reflective IR Sensor	OPB732	5 V emitter circuit	1	50.00	50.00	mA
	N/A	N/A	N/A	0	0.00	0.00	mA

+12V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	DC Gearmotor	FIT0492-A	12 V	1	2,190.00	2,190.00	mA
	H-Bridge Motor Driver	DRV8876PWPR	4.5-37 V	1	7.00	7.00	mA
						0.00	mA
					Subtotal	2,197.00	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	2,746.25	mA
Source	12 V Wall Supply	GST60A12-P1J	12 V regulated output	1	5,000.00	5,000.00	mA
					Total Remaining Current Available on +12V Rail	2,253.75	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano	DM164150	1.8-5.5 V	1	100.00	100.00	mA
	Quad Op-Amp	MCP6004-I/P	1.8-6 V	1	0.68	0.68	mA
	Reflective IR Sensor + IR LED	OPB732	5 V emitter circuit	1	50.00	50.00	mA
					Subtotal	150.68	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	188.35	mA
Regulator	+5V Linear Regulator	L7805CV	7-35 V input	1	1,500.00	1,500.00	mA
					Total Remaining Current Available on +5V Rail	1,311.65	mA

Unused Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
						0.00	mA
						0.00	mA
					Subtotal	0.00	mA
					Safety Margin	25%	
					Total Current Required on Unused Rail	0.00	mA

Regulator	None	N/A	N/A	1	0.00	0.00	mA
					Total Remaining Current Available on Unused Rail	0.00	mA

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	12 V Wall Supply	GST60A12-P1J	100-240 VAC	12 V	5,000.00	5,000.00	mA

Power Rails Connected to External Power Source 1	+12V Motor Rail	Direct from 12 V supply	12 V	12 V	2,746.25	2,746.25	mA
	+5V Linear Regulator	L7805CV	12 V input	5 V	188.35	188.35	mA
	Unused Rail	N/A	N/A	N/A	0.00	0.00	mA
					Total Remaining Current Available on External Power Source 1	2,065.40	mA

Note: Section D uses each rail's required current (including the 25% safety margin), not the regulator's maximum output-current rating. The L7805 input-current estimate is conservatively approximated by the 5 V rail load; regulator quiescent current is small relative to the available margin.

External Power Source 2 (Not Used)	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 2 Selection	None	N/A	N/A	N/A	0.00	0.00	mA

Power Rails Connected to External Power Source 2	None	N/A	N/A	1	0.00	0.00	mA
					Total Remaining Current Available on External Power Source 2	0.00	mA

Calculated Battery Life (Not Applicable)							
	Component Name	Part Number	Supply Voltage Range		Capacity (mAh)	Required By Regulators	
	No Battery Used	N/A	N/A		0.00	0.00	
					Battery Life	0.00	hours

Notes:

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).

If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit